

Program: Diploma in Integrated Circuit Design and Fabrication	
Course Code: 5371	Course Title: Microcontroller and Embedded System
Semester: 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- Acquire knowledge about microcontrollers embedded processors and their applications.
- Introduce the importance of embedded systems
- Introduce ATMEGA 32 microcontroller and its suitability for embedded systems
- Familiarize embedded C programming
- Familiarize serial communication and data transfer between embedded system

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Engineering mathematics principles and theorems	1002 & 2002	Mathematics I & II	1 & 2
Different number systems, codes, combinational and sequential circuits, memory	3044	Digital Electronics	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Explain the basics of embedded systems and its architecture	15	Understanding
CO2	Explain the architecture and functionality of 8051 microcontroller	14	Understanding
CO3	Make use of AVR Microcontrollers to develop embedded programs using embedded C	15	Applying

CO4	Make use of AVR microcontroller to interface various peripheral devices using embedded and Familiarize RTOS	14	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	3		3				
CO4	3		3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basics of embedded systems and its architecture		
M1.01	Distinguish between embedded system & general-purpose computer	1	Understanding
M1.02	Classify embedded systems and explain its application areas	2	Understanding
M1.03	Explain Hardware Architecture of embedded system	6	Understanding
M1.04	Explain Software Architecture of embedded system	4	Understanding
M1.05	Explain Real Time Operating Systems	2	Understanding
Contents: Embedded Systems - Definition, difference from general purpose computers - Classification of embedded systems, Application areas. Hardware Architecture of embedded system – Building blocks of an embedded system, Core of embedded system – categories, Memory –ROM and RAM, Sensors and I/O subsystems (LED, Relay, Stepper motor). On board (I2C, SPI, UART) and external communication interfaces (RS 232, USB, Bluetooth, Wi-Fi). Software Architecture of embedded system -Software Layers. Kernel, types of operating			

systems- GPOS, RTOS. Real time operating systems- Tasks, process, threads, multiprocessing and multi-tasking List popular Real Time Operating Systems (any 10)			
CO2	Explain the architecture and functionality of 8051 microcontroller		
M2.01	List the features of 8051 microcontroller.	2	Understanding
M2.02	Explain the block diagram of 8051 microcontroller	2	Understanding
M2.03	Explain memory organization of 8051.	6	Understanding
M2.04	Explain concept of interrupts, types, priority, registers for interrupts and handling of interrupts.	4	Understanding
	Series Test – I	1	
Contents: Introduction to Microcontroller, simple block diagram of 8051, Data memory organization, program memory organization, PSW register. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register.			
CO3	Make use of AVR Microcontrollers to develop embedded programs using embedded C		
M3.01	Familiarize AVR controllers' family members and criteria to select a microcontroller.	2	Understanding
M3.02	Explain block diagram of Atmega32.	3	Understanding
M3.03	Illustrate Registers, Memory organization, Status register, Program counter, I/O ports and its registers, Interrupts, priority of interrupts	3	Understanding
M3.04	Illustrate Timers in AVR	3	Understanding
M3.05	Develop embedded C programs for logic operations, data conversions and I/O operations.	3	Applying
M3.06	Compare 8051 and Atmega 32 Microcontroller.	1	Understanding
Contents: AVR Microcontroller Architecture - Comparison of AVR family members and Selection of a microcontroller, ATmega32- Simplified Block diagram of ATmega32 microcontroller - Registers, - data memory - I/O memory -SFRs - internal data SRAM, Status register, Program Counter and Program ROM space, I/O ports, Registers associated with I/O ports, Timers- 0,1,2 (Block level) associated registers, Interrupts. AVR programming using embedded C -Data types, I/O programming, logic operations - data conversion programs - time delays- programming of timers 0 - timer 1 - timer2. Comparison of 8051 and Atmega 32 Microcontroller.			

CO4	Make use of AVR microcontroller to interface various peripheral devices using embedded C.		
M4.01	Outline the need for interfacing, and types of interfacing devices.	1	Understanding
M4.02	Illustrate the interfacing of LED, Push button, Relay, Sensors and Seven segment Display with AVR	3	Applying
M4.03	Make use of AVR to interface LCD and Keyboard	3	Applying
M4.04	Make use of AVR microcontroller to interface DC motor, Servo motor and stepper motor and ADC	3	Applying
M4.05	Implement I2C interfacing with AVR	3	Applying
M4.06	Understand SPI interface with AVR	1	Understanding
	Series Test – II	1	
Contents: Interfacing of LED, Push button, Relay, Sensors (Temperature sensor, IR sensor) Seven segment Display, LCD, Keyboard Interfacing, Motor (DC, Servo, Stepper), ADC interfacing. On-board communication interfaces with AVR – basics of I2C interfacing (like real time clock interfacing) and SPI interfacing with AVR			

Text/Reference

T/R	Book Title/Author
T1	Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller and Embedded Systems Using Assembly and C - Pearson Education- Second Edition.
T2	Rajkamal, Embedded Systems Architecture, Programming and Design, TMH, 2003
T3	K.V. Shibu, Introduction to Embedded Systems, 2e, McGraw Hill Education India, 2016.
T4	KVK K Prasad
R1	Michael J. Pont, Embedded C, Pearson Education, Second Edition

Online resources:

Sl.No.	Website Link
1	https://www.electronicshub.org
2	https://www.elprocus.com
3	https://www.tutorialspoint.com
4	https://www.studyelectronics.in/embedded-programming-tutorial-chapter-1-beginners/
5	https://www.tutorialspoint.com/embedded_systems/index.htm
6	https://embeddedschool.in/avr-microcontroller-programming/
7	https://learn.sparkfun.com/tutorials/i2c/all
8	https://www.best-microcontroller-projects.com/i2c-tutorial.html
9	https://www.electronicwings.com/avr-atmega/atmega1632-i2c
10	https://www.electronicwings.com/avr-atmega/atmega1632-spi
11	https://www.intel.com/content/www/us/en/docs/programmable/683525/21-3/overview-of-the-microc-os-ii-rtos.htm